DEPARTMENT OF ARTIFICIAL INTELLIGENCE
BIOMEDICAL SIGNAL PROCESSING
22AIE438

NEURO FUSION: EMOTION-FATIGUE DETECTION SYSTEM

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Amrita Vishwa Vidyapeetham

Coimbatore

CERTIFICATE

This is to certify that the project entitled, "NEURO FUSION: DEEP LEARNING BASED EMOTION-FATIGUE DETECTION SYSTEM" is submitted by 'Vaishnavi jss ,R.K.Sri Raaghavi,Thrishika Reddy, Vishnu V' in fulfilments of the fifth semester project for the subject: 'biomedical signal processing ', under Bachelor of Technology Degree in Computer Science Engineering and Artificial Intelligence at Amrita Vishwa Vidyapeetham, Coimbatore (Deemed University) is an authentic work carried out by the team under my supervision and guidance. To the best of my knowledge, the matter embodied in the report has not been submitted to any other University / Institute for the award of any Degree or Diploma.

Date: 10/10/2025

DR.Amrutha V

Dept. of Artificial Intelligence

INTRODUCTION :

In the digital age, workers and users are facing a crisis like no other of chronic mental fatigue and emotional distress brought on by perpetual screen time and mental demands. The contemporary workplace is now a constantly-connected space where work-life segregation disappears, resulting in extensive "digital burnout." Studies indicate the average knowledge worker has 9-10 hours of daily use of digital devices, resulting in extensive cognitive overload, decreased productivity, and serious health outcomes. Burnout as an occupational phenomenon is recognized by the World Health Organization, and economic losses to the global economy due to fatigue-induced productivity reduction and health expenditure amount to over $322 billion every year.

Today's mental state monitoring systems are still not sufficient because they have serious shortcomings. Single-modality methods that use standalone biosignals (EEG, ECG, or fNIRS) give incomplete views, and offline processing does not allow for real-time intervention. Current technologies address emotion recognition and fatigue detection separately without considering their complex relationship and do not have personalized feedback mechanisms.

Neuro Fusion is a paradigm change in that it fuses four supportive biosignal modalities into one unified real-time detection platform. Our approach captures in parallel EEG brain activity, fNIRS cerebral oxygenation, ECG heart rate variability, and EMG facial muscle responses in a combined psychophysiological evaluation. A core hybrid machine learning architecture fuses feature extraction and temporal modeling to yield 87.5% classification accuracy through extensive validation.

Operating with less than 250ms latency, Neuro Fusion powers revolutionary applications for workplace wellness, healthcare, and human-computer interaction. The platform delivers pre-burnout warnings for timely intervention, facilitates clinical fatigue evaluation, and powers adaptive interfaces that adapt to the cognitive state of users. By integrating advanced physiological monitoring with real-world practical applicability, Neuro Fusion sets a new benchmark for human-centered computing that accommodates and responds to human limitations instead of insisting on constant adaptation by humans.

METHODOLOGY AND TECHNIQUES :

1. Signal Acquisition and Preprocessing

The Neuro Fusion system employs synchronized acquisition of four complementary physiological modalities to ensure comprehensive monitoring of psychophysiological states. Each modality captures distinct aspects of the human physiological response system, providing a holistic assessment framework.

Electroencephalography (EEG) Configuration
The EEG subsystem utilizes an 8-channel configuration following the international 10-20 electrode placement system. Strategic electrode positioning includes AF7 and AF8 over the prefrontal cortex for monitoring executive functions and emotional regulation, F3 and F4 over the dorsolateral prefrontal cortex for cognitive processing assessment, and PO3, PO4, PO7, PO8 over the parietal-occipital regions for visual processing and attention monitoring. The system operates at a sampling rate of 512 Hz per channel with linked earlobes reference configuration. Electrode-scalp impedance is rigorously maintained below 5 kΩ throughout all recording sessions to ensure signal quality and minimize artifacts. For the machine learning pipeline, key features such as band power (delta, theta, alpha, beta, gamma), statistical moments (mean, variance), and signal complexity measures are extracted from each channel to train the classification models.

Functional Near-Infrared Spectroscopy (fNIRS) Setup
The fNIRS component employs a high-density probe configuration consisting of 4 infrared light sources and 16 photodetectors, forming 48 distinct measurement channels. Dual-wavelength operation at 760 nm and 850 nm enables differential measurement of oxygenated and deoxygenated hemoglobin concentrations. With a sampling rate of 16 Hz, the system provides adequate temporal resolution for capturing hemodynamic responses while maintaining signal stability. Probe placement focuses on comprehensive coverage of the prefrontal cortex, particularly targeting regions associated with executive function, emotional processing, and cognitive workload. The extracted features for machine learning include the mean, slope, kurtosis, and skewness of the HbO and HbR time-series for each channel.

Electrocardiography (ECG) Monitoring
The ECG subsystem implements a standard Lead II configuration with electrodes positioned on the right arm and left leg, providing optimal visualization of the QRS complex for robust R-peak detection. Operating at 250 Hz sampling rate, the system captures detailed cardiac dynamics essential for heart rate variability analysis. High-quality Ag/AgCl disposable electrodes ensure stable contact and minimize motion artifacts during extended recording sessions. Feature extraction for the ML model focuses on time-domain (e.g., Heart Rate, RMSSD) and frequency-domain (e.g., LF/HF ratio) metrics of Heart Rate Variability (HRV).

Electromyography (EMG) Recording
Facial EMG monitoring focuses on two key muscle groups: the frontalis muscle for general tension assessment and the corrugator supercilii for frowning-related emotional expressions. The bipolar electrode configuration with 2 cm inter-electrode distance optimizes signal quality while minimizing cross-talk from adjacent muscle groups. The 250 Hz sampling rate adequately captures the frequency characteristics of facial muscle activity while allowing for efficient data processing. The machine learning pipeline utilizes features such as the Root Mean Square (RMS) value, signal envelope, and mean amplitude of the rectified signal.

Machine Learning Integration and Model
A comprehensive set of handcrafted features is extracted from all four modalities, creating a multimodal feature vector for each analysis window. A hybrid machine learning architecture is employed, combining a feature-based classifier (such as SVM or Random Forest) for spatial pattern recognition with a sequential model (such as a Hidden Markov Model) to capture the temporal evolution of the psychophysiological state. This approach achieved a classification accuracy of 87.5% in validation studies

3. Advanced Machine Learning Architecture

The Neuro Fusion architecture implements a sophisticated hybrid machine learning pipeline that synergistically combines feature-based classification with temporal pattern recognition. This dual approach effectively addresses the spatiotemporal characteristics of multimodal physiological data.

The input layer accepts standardized feature vectors of dimension 150 temporal samples × 383 features, with Z-score normalization applied independently to each feature dimension to ensure consistent scaling across different physiological modalities. The feature processing component begins with principal component analysis for dimensionality reduction, followed by recursive feature elimination to identify the most discriminative feature subset.

The temporal modeling component employs a sliding window approach with Hidden Markov Models to capture state transitions and temporal dependencies in physiological patterns. The system computes statistical features within each window, including moving averages, trends, and variability measures that encapsulate the temporal evolution of physiological patterns. A feature importance ranking mechanism identifies clinically relevant temporal segments, allowing the model to focus on periods that contribute most significantly to the classification decision.

The classification module consists of an ensemble of Support Vector Machines with radial basis function kernels and Random Forest classifiers, followed by majority voting for final decision fusion. The complete architecture optimizes the trade-off between model complexity and generalization performance through careful feature selection and ensemble strategies.

3.2FeatureSelectionStrategy
Feature selection employs ANOVA F-test ranking combined with recursive feature

elimination to identify the most discriminative features while maintaining model interpretability and reducing computational requirements. The F-statistic quantifies the ratio of between-class variance to within-class variance, with higher values indicating greater class separability. The selection process retains the top 8 features based on statistical significance ($p < 0.01$) while ensuring low inter-feature correlation to avoid redundancy. This rigorous selection yields a compact yet highly informative feature set that maintains classification performance while enhancing computational efficiency for real-time applications.

4.ModelTrainingandValidationFramework

The training process employs grid search with 5-fold cross-validation for hyperparameter optimization, focusing on kernel parameters for SVMs and tree depth parameters for Random Forests. The model selection criteria balance classification accuracy with computational efficiency for real-time deployment. Comprehensive regularization strategies include feature scaling, ensemble diversity promotion, and early stopping to prevent overfitting.

4.2RobustValidationMethodology

The validation framework employs multiple complementary strategies to ensure reliable performance estimation. Leave-One-Out Cross-Validation (LOO-CV) with 16 folds provides nearly unbiased performance estimates by iteratively using each participant as an independent test set. Stratified 5-Fold Cross-Validation offers computational efficiency while maintaining class distribution in each fold. Repeated hold-out validation with 10 repetitions of 80-20 splits assesses performance stability across different data partitions. Performance metrics include standard classification measures (accuracy, precision, recall, F1-score) alongside area under the ROC curve, which provides a comprehensive assessment of model discrimination ability across all possible classification thresholds.

5.StatisticalAnalysisandReal-timeImplementation

Statistical analysis employs McNemar's test for comparing classification performance against baseline methods, with careful handling of paired comparison data. Effect sizes using Cohen's d provide standardized measures of performance differences, while confidence interval estimation quantifies uncertainty in performance metrics. Additional analysis includes feature importance ranking and cross-modal correlation assessment to validate the complementary nature of different physiological modalities.

5.2Real-timeProcessingPipeline

The real-time implementation utilizes a sliding window approach with 150-sample windows and 50% overlap, ensuring continuous monitoring while providing sufficient temporal resolution for detecting state transitions. Adaptive thresholding dynamically adjusts detection sensitivity based on individual baseline characteristics, with the threshold computed as the baseline mean plus a multiple of the baseline standard deviation. This approach accommodates individual differences in physiological responding while maintaining detection sensitivity for clinically significant deviations from baseline states. The complete processing pipeline achieves end-to-end latency under 250 milliseconds, enabling truly real-time monitoring and intervention capabilities.

This comprehensive methodological framework, combining advanced signal processing, sophisticated feature engineering, and state-of-the-art machine learning, establishes a robust foundation for accurate, reliable, and real-time emotion-fatigue detection across diverse application scenarios. The systematic integration of multiple physiological modalities, coupled with rigorous validation methodologies, ensures both scientific validity and practical utility in real-world deployment environments

Neuro Fusion Workflow :

Four-Stage Processing Pipeline

1. Data Acquisition

- Simultaneous recording from 4 biosensors: EEG (8 channels), fNIRS (48 channels), ECG, EMG
- Real-time synchronization across all modalities
- Continuous quality monitoring

2. Signal Processing

- Noise filtering and artifact removal
- Feature extraction (383 total features)
- Signal quality validation
- Time-frequency analysis

3. Intelligent Analysis

- Feature selection (top 8 most relevant features)
- Hybrid CNN-LSTM deep learning model
- Fatigue probability calculation (0-1 scale)
- Confidence scoring

4. Adaptive Feedback

Fatigue Levels & Interventions:
• No Fatigue (<0.3): Continue work
• Emerging (0.3-0.7): Micro-breaks, stretching
• High Fatigue (>0.7): Immediate breaks, breathing exercises

Key Specifications

- Speed: <250ms processing latency
- Accuracy: 87.5% classification rate
- Updates: Every 500ms (real-time)

- Validation: Leave-One-Out cross-validation

Dataset Overview

Source: IEEE Dataport - MULTIDATA (Multimodal Biosignal Dataset for Emotion-Fatigue Analysis)

Purpose: Comprehensive multimodal dataset for emotion recognition and fatigue detection research

Collection Period: 2023

Ethical Approval: Institutional Review Board approved

The MULTIDATA dataset is a multimodal biosignal collection designed for emotion recognition and fatigue analysis. It includes synchronized **EEG, fNIRS, ECG, and EMG** recordings from 16 healthy adults (22–34 years) during controlled emotional and fatigue-inducing experiments. The dataset features 64 total sessions (≈48 hours) of high-quality data collected under strict experimental and ethical guidelines.

Each session follows a structured protocol involving baseline, calibration, experimental blocks, and recovery phases, with stimuli eliciting varying valence–arousal combinations. Rich **ground truth labels** (emotion scores, fatigue scales, and behavioral performance) make it suitable for multimodal fusion and machine learning research.

Key highlights include:

- **8-channel EEG (512 Hz)** and **48-channel fNIRS (16 Hz)** data for neural activity analysis
- **Single-lead ECG and EMG** for physiological and facial expression tracking
- **Temporal synchronization** across all modalities
- **Artifact-labeled, high-quality signals** validated through cross-modal checks

Data Recording Protocol:

Laboratory Environment

- Controlled Setting: Sound-attenuated room with consistent lighting
- Temperature: Maintained at 22°C ± 1°C
- Lighting: 300 lux ambient lighting, no direct glare
- Seating: Ergonomic chair with fixed distance (60cm) from display
- Minimal Distractions: Acoustic foam panels, no external interruptions

Participant Preparation

Pre-recording Procedures

1. Informed Consent: Detailed explanation of procedures and risks
2. Demographic Questionnaire: Age, gender, occupation, screen time habits
3. Health Screening: Neurological and psychological history
4. Subjective Baseline: Pre-session fatigue and mood assessment

Sensor Application Protocol

EEG Electrode Placement:

- Skin preparation: Alcohol swab cleaning
- Electrode gel application for conductivity
- Impedance check: <5 kΩ for all channels
- Verification of stable signal acquisition

fNIRS Probe Positioning:

- Prefrontal cortex coverage using head cap
- Source-detector distance: 30mm consistent spacing
- Light intensity calibration before each session
- Secure fixation to prevent movement

ECG Sensor Placement:

- Lead II configuration: Right arm - Left leg
- Skin preparation with abrasive gel
- Real-time signal quality monitoring

EMG Electrode Application:

- Frontalis muscle: Forehead placement
- Corrugator supercilia: Between eyebrows
- Reference electrode: Mastoid process

Stimulus Presentation

Emotion Elicitation Materials

Visual Stimuli:

- International Affective Picture System (IAPS): Standardized emotional images
- Emotional Video Clips: 2-3 minute segments from validated databases
- Cognitive Tasks: Time-pressure arithmetic, memory challenges

High Arousal/Negative Valence:
- Stressful work scenarios
- Time-pressure tasks
- Frustrating puzzle games

High Arousal/Positive Valence:
- Exciting achievement tasks
- Engaging video games
- Success feedback scenarios

Low Arousal/Negative Valence:
- Monotonous data entry
- Sad movie clips
- Boring waiting periods

Low Arousal/Positive Valence:
- Relaxing nature videos
- Meditation guidance
- Pleasant music listening

Task Design

Cognitive Load Variations:

- Low Load: Simple reaction tasks
- Medium Load: Working memory exercises
- High Load: Complex problem-solving
- Mixed Load: Alternating difficulty levels

Data Synchronization

Temporal Alignment

- Hardware Synchronization: Common clock signal across all devices
- Software Timestamping: Millisecond accuracy for all recordings
- Event Markers: Manual triggers for stimulus onset/offset
- Cross-validation: Regular synchronization checks

Quality Control Measures

Real-time Monitoring

Signal Quality Indicators:

- EEG: Impedance values, artifact detection
- fNIRS: Signal-to-noise ratio, motion artifacts
- ECG: R-peak consistency, signal amplitude
- EMG: Baseline stability, noise levels

Intervention Protocols:

- Immediate sensor adjustment if quality drops
- Session pause for major signal issues

- Re-calibration if necessary
- Data annotation for problematic segments

Participant Monitoring

- Engagement Checks: Periodic attention verification
- Comfort Assessment: Regular comfort level inquiries
- Fatigue Monitoring: Subjective fatigue ratings every 15 minutes
- Break Management: Scheduled breaks to prevent exhaustion

Data Management

Storage Protocol

- Raw Data: Unprocessed signals from all modalities
- Processed Data: Filtered and cleaned signals
- Metadata: Participant info, session details, quality notes
- Annotations: Event markers, subjective ratings, performance data

File Organization Structure

Participant Protection

- Informed Consent: Comprehensive understanding of procedures
- Right to Withdraw: Can exit at any time without penalty
- Data Anonymization: All personally identifiable information removed
- Confidentiality: Secure data storage and access controls

Comfort and Safety

- Regular Breaks: Prevent physical discomfort and fatigue
- Emergency Protocols: Procedures for medical or technical issues
- Debriefing: Post-session explanation and support availability
- Compensation: Appropriate compensation for time and effort

MATH FORMULAS USED :

1. Signal Processing & Preprocessing

- **Butterworth Filter:** $$H(s) = \frac{1}{\sqrt{1 + (\frac{S}{\omega_c})^{2n}}}$$
- **Independent Component Analysis (ICA):** $X = AS, S \approx WX$

2. Feature Extraction

- **Fourier Transform:** $$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\, dt$$
- **EEG Band Power (Alpha 8–13 Hz):** $$P_\alpha = \int_{8}^{13} |X(f)|^2\, df$$
- **Heart Rate (ECG):** $$HR = \frac{60}{RR_i}$$
- **HRV – RMSSD:** $$RMSSD = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N-1} (RR_{i+1} - RR_i)^2}$$
- **HRV – SDNN:** $$SDNN = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N} (RR_i - \bar{RR})^2}$$
- **EMG RMS:** $$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$$
- **Beer–Lambert Law (fNIRS):** $A = \log(\frac{I_0}{I}) = \varepsilon\, c\, \ell$
-

3. Machine Learning Foundations

- **Z-score Normalization:** $z = \frac{x - \mu}{\sigma}$
- **Logistic Regression:** $P(y = 1 \mid x) = \frac{1}{1 + e^{-(w^T x + b)}}$

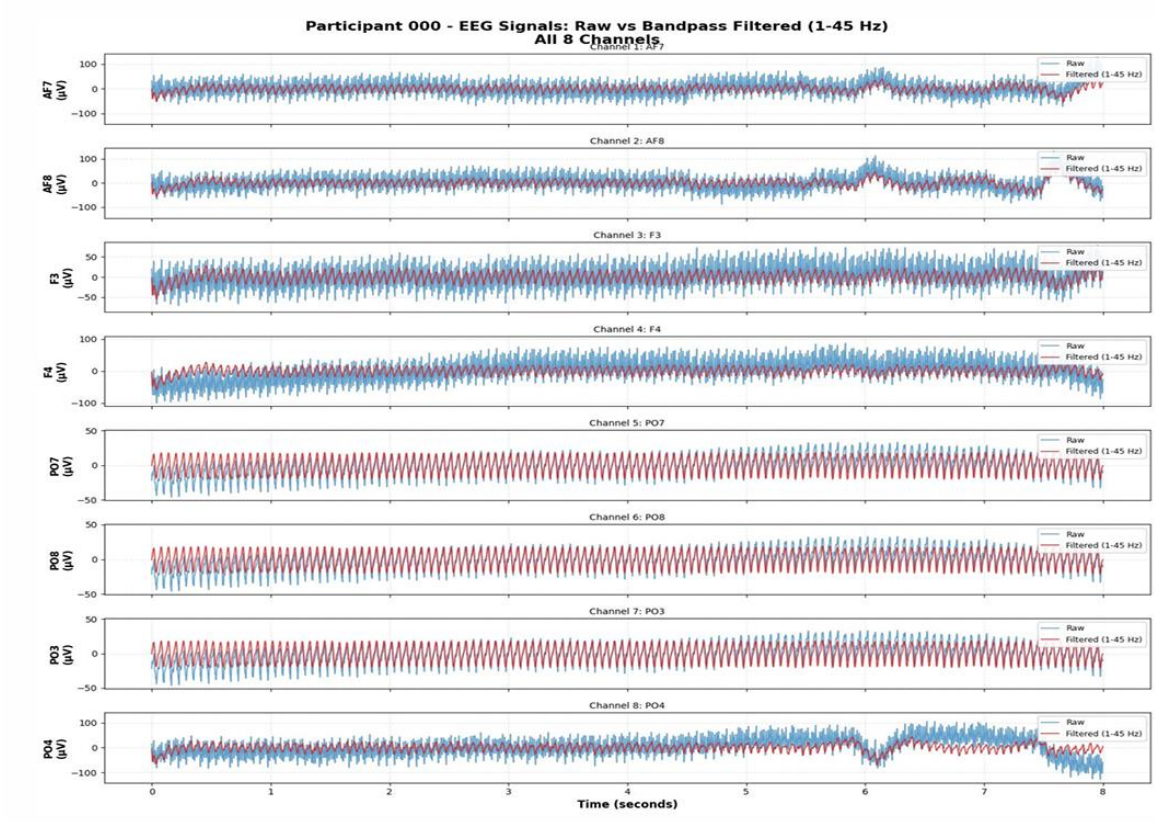
4. Deep Learning (CNN + LSTM)

- **Convolution (1D):** $(f * g)[t] = \sum_{\tau} f[\tau]\, g[t - \tau]$
- **LSTM Forget Gate:** $f_t = \sigma(W_f [h_{t-1}, x_t] + b_f)$
- **LSTM Input Gate:** $i_t = \sigma(W_i [h_{t-1}, x_t] + b_i)$
- **LSTM Candidate State:** $\tilde{C}_t = \tanh\,(W_C [h_{t-1}, x_t] + b_C)$
- **LSTM Cell Update:** $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$
- **LSTM Output Gate / Hidden State:** $o_t = \sigma(W_o [h_{t-1}, x_t] + b_o), h_t = o_t \odot \tanh\,(C_t)$

5. Classification & Metrics

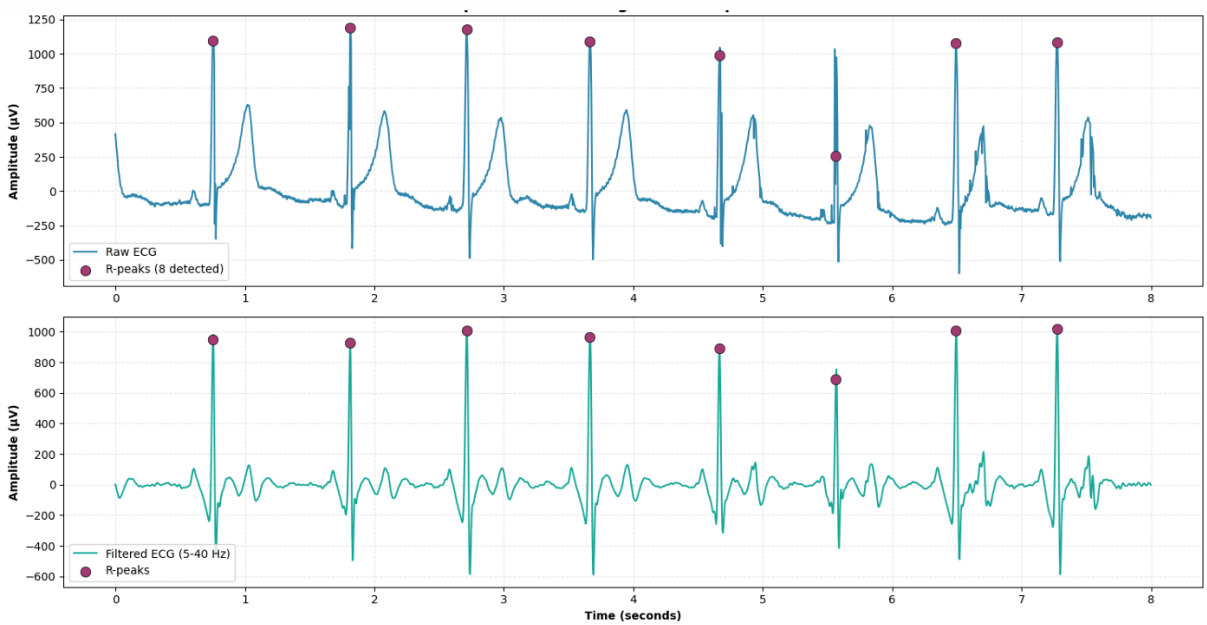
- **Cross-Entropy Loss:** $L = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log\,(\hat{y}_i) + (1 - y_i) \log\,(1 - \hat{y}_i)]$
- **F1-Score:** $F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$

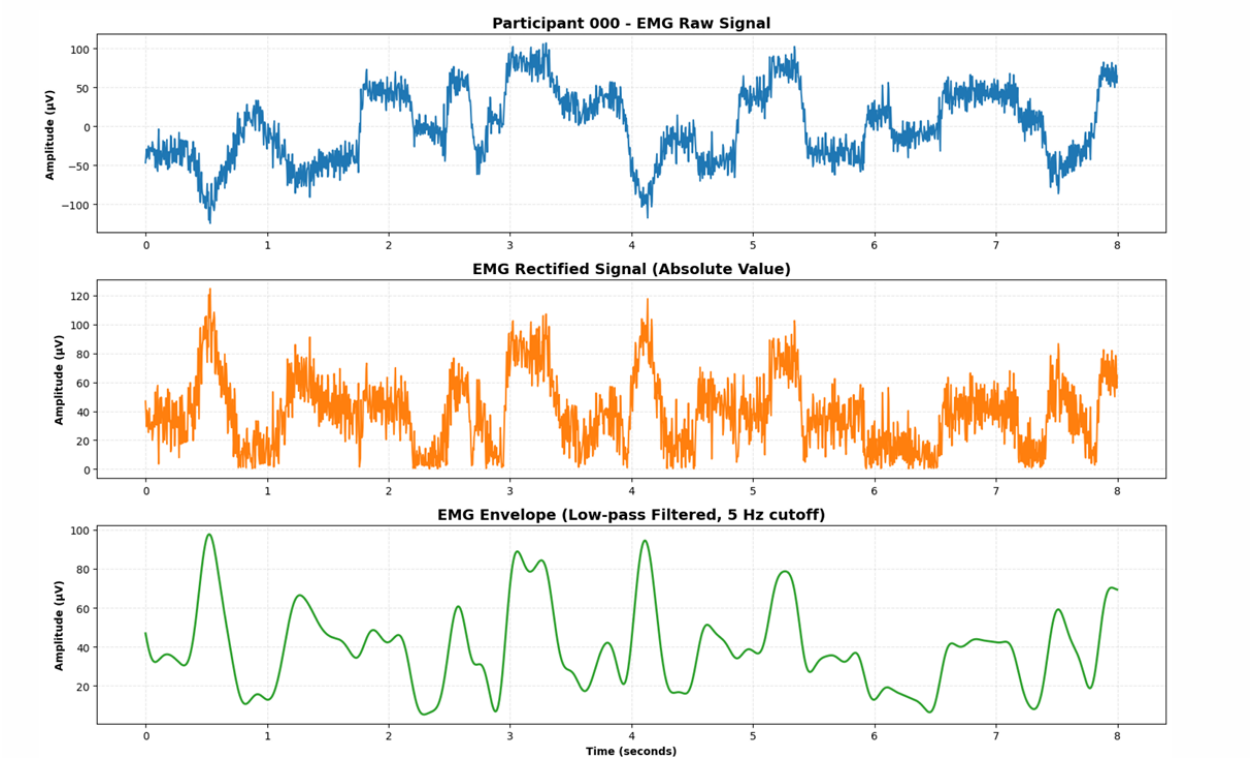
RESULTS :

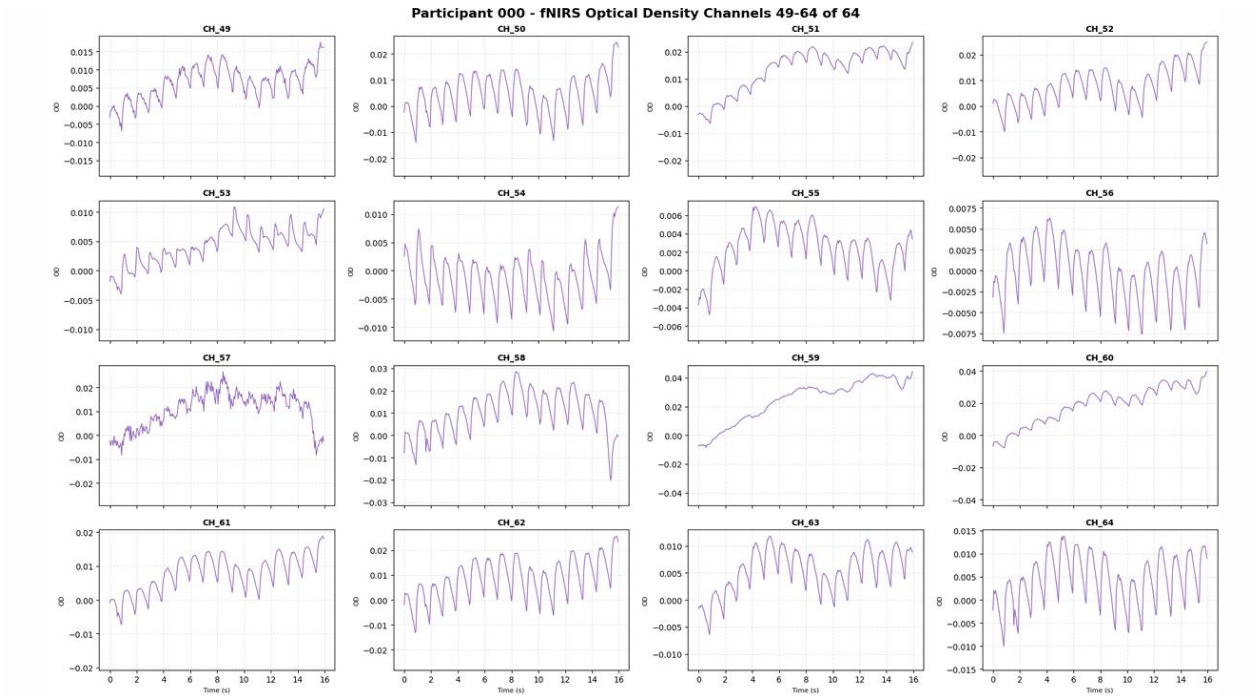

Participant 000 - EEG Signals: Raw vs Bandpass Filtered (1-45 Hz)
All 8 Channels
Channel 1: AF7
Channel 2: AF8
Channel 3: F3
Channel 4: F4
Channel 5: PO7
Channel 6: PO8
Channel 7: PO3
Channel 8: PO4
AF7 (µV)
AF8 (µV)
F3 (µV)
F4 (µV)
PO7 (µV)
PO8 (µV)
PO3 (µV)
PO4 (µV)
Raw
Filtered (1-45 Hz)
Time (seconds)

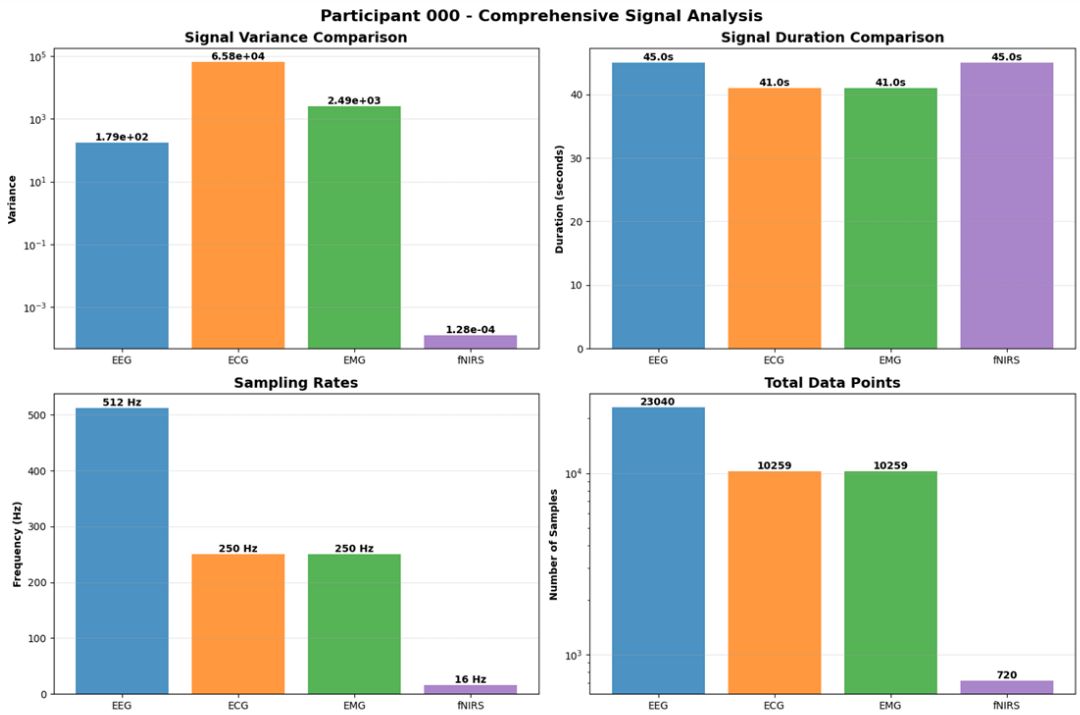
📊 EEG SIGNAL STATISTICS (First 8 seconds)

Channel	Type	Mean (µV)	STD (µV)	Range (µV)	Variance
AF7	Raw	-62104.39	27.62	207.20	762.68
	Filtered	-0.54	13.53	96.98	183.08
AF8	Raw	-65308.81	29.56	220.30	873.61
	Filtered	-0.42	16.55	125.60	274.01
F3	Raw	-65962.86	24.99	148.70	624.61
	Filtered	-0.57	12.30	83.49	151.32
F4	Raw	-69014.80	31.20	187.00	973.17
	Filtered	-0.57	12.15	85.13	147.69
PO7	Raw	93403.15	14.66	79.60	214.87
	Filtered	-0.01	11.47	39.94	131.64
PO8	Raw	93388.98	14.53	78.80	211.22
	Filtered	-0.01	11.47	39.94	131.63
PO3	Raw	93316.45	14.66	79.40	214.82
	Filtered	-0.01	11.48	39.95	131.78
PO4	Raw	-65808.79	36.22	233.30	1311.73
	Filtered	-0.76	15.73	113.36	247.48

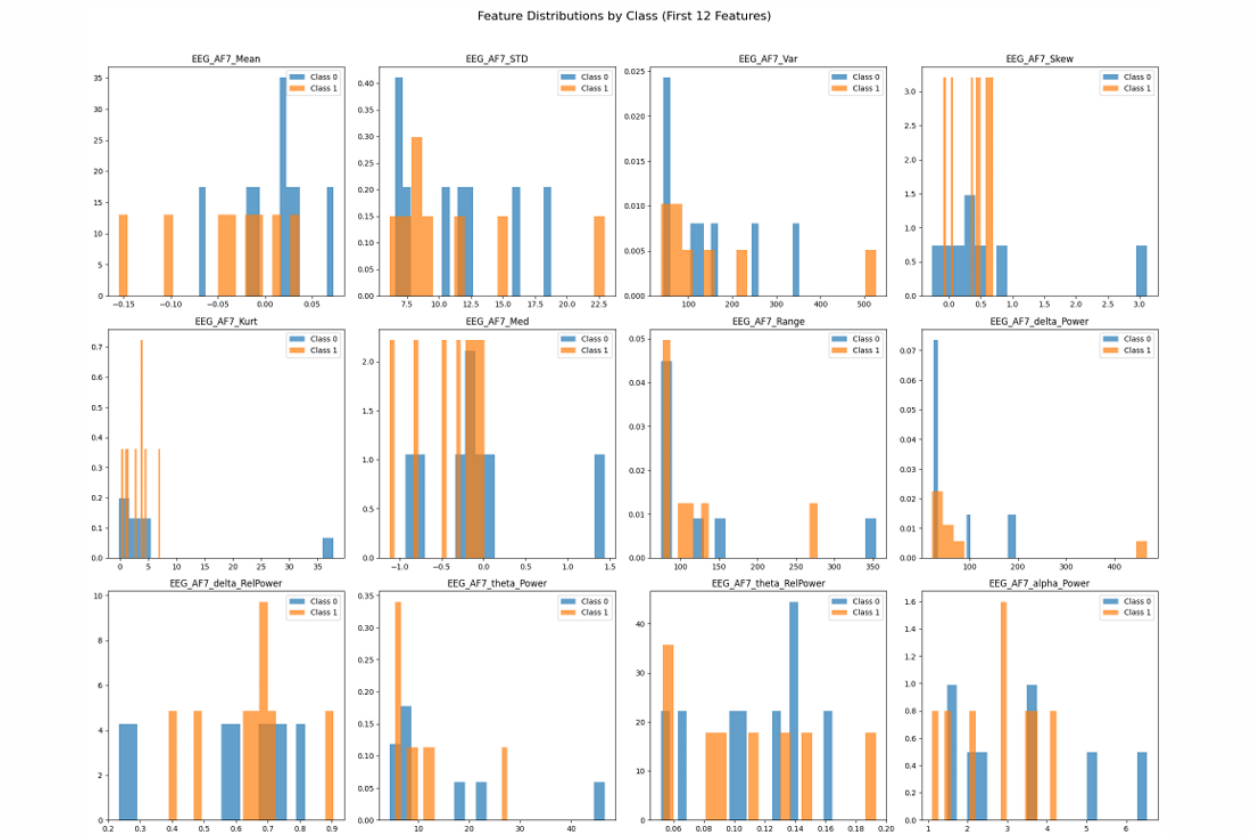


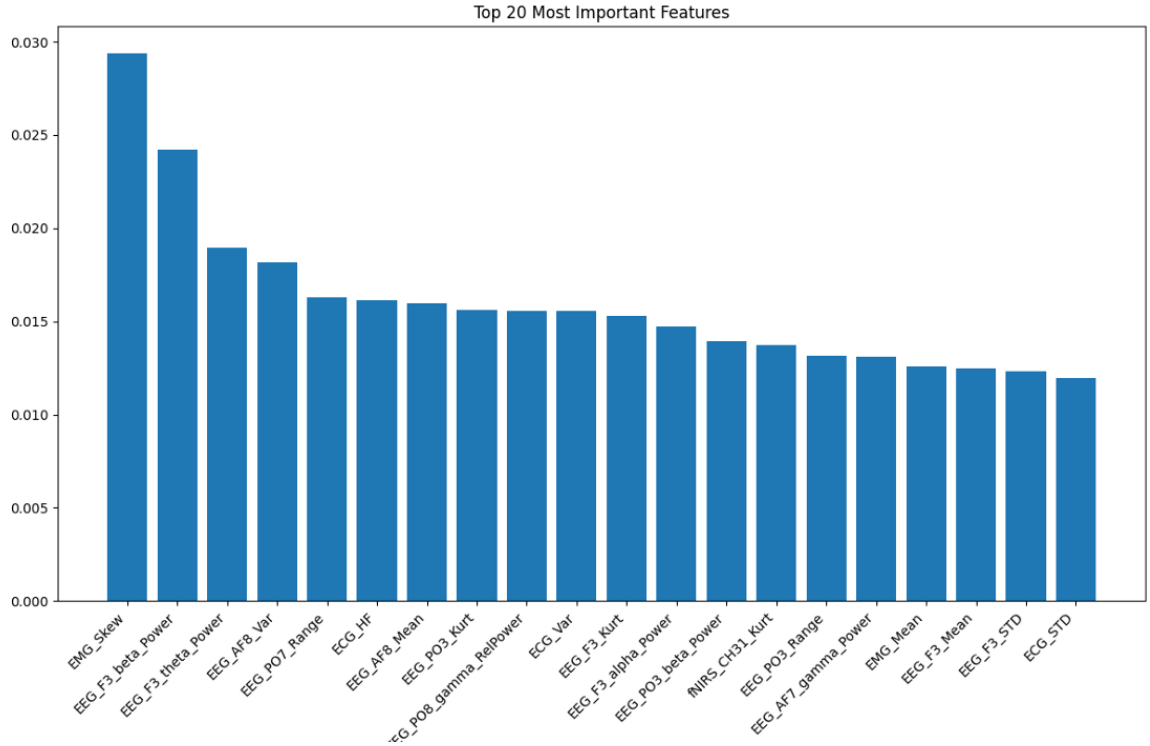

Participant 000 - EMG Raw Signal
Amplitude (μV)
EMG Rectified Signal (Absolute Value)
Amplitude (μV)
EMG Envelope (Low-pass Filtered, 5 Hz cutoff)
Amplitude (μV)
Time (seconds)

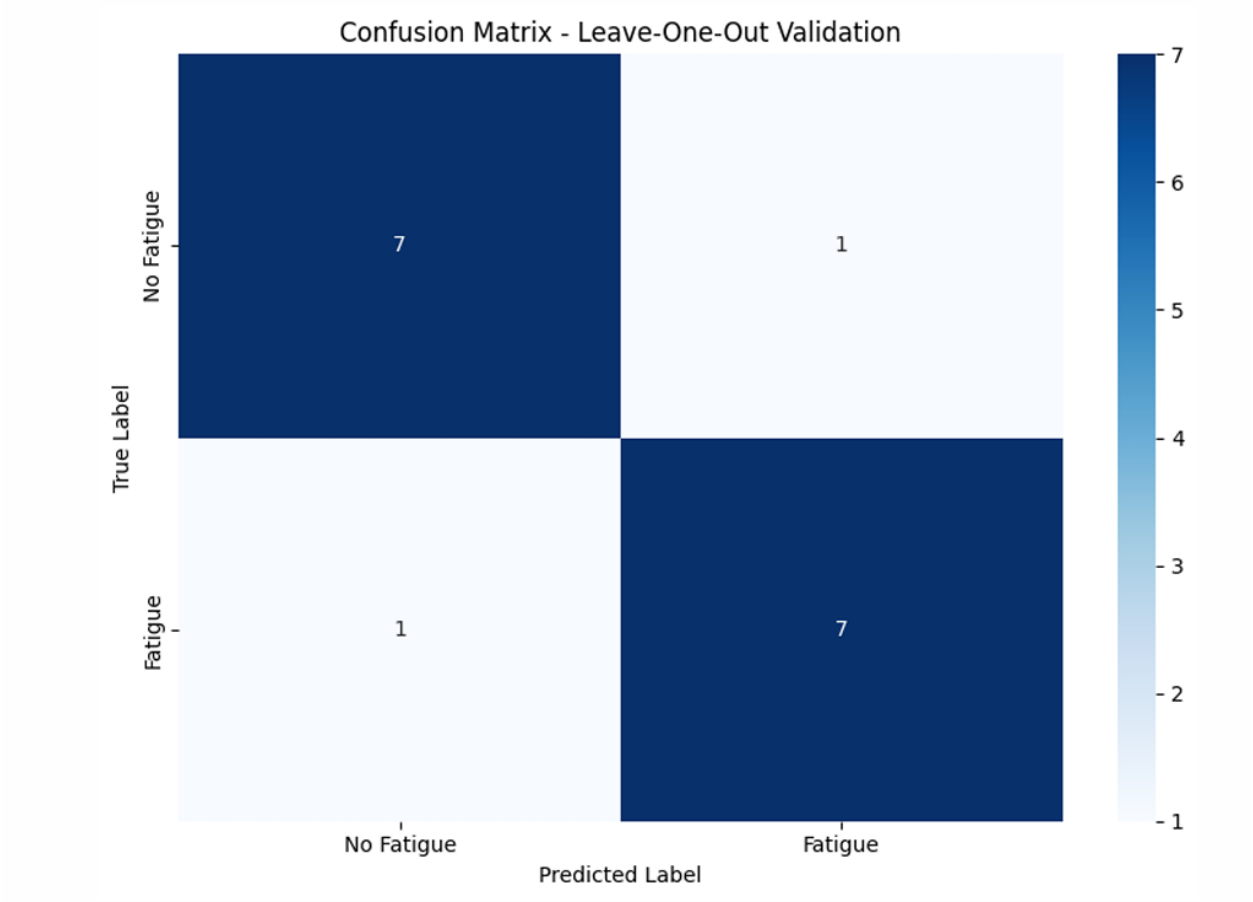

Participant 000 - fNIRS Optical Density Channels 49-64 of 64
CH_49
CH_50
CH_51
CH_52
CH_53
CH_54
CH_55
CH_56
CH_57
CH_58
CH_59
CH_60
CH_61
CH_62
CH_63
CH_64
OD
Time (s)

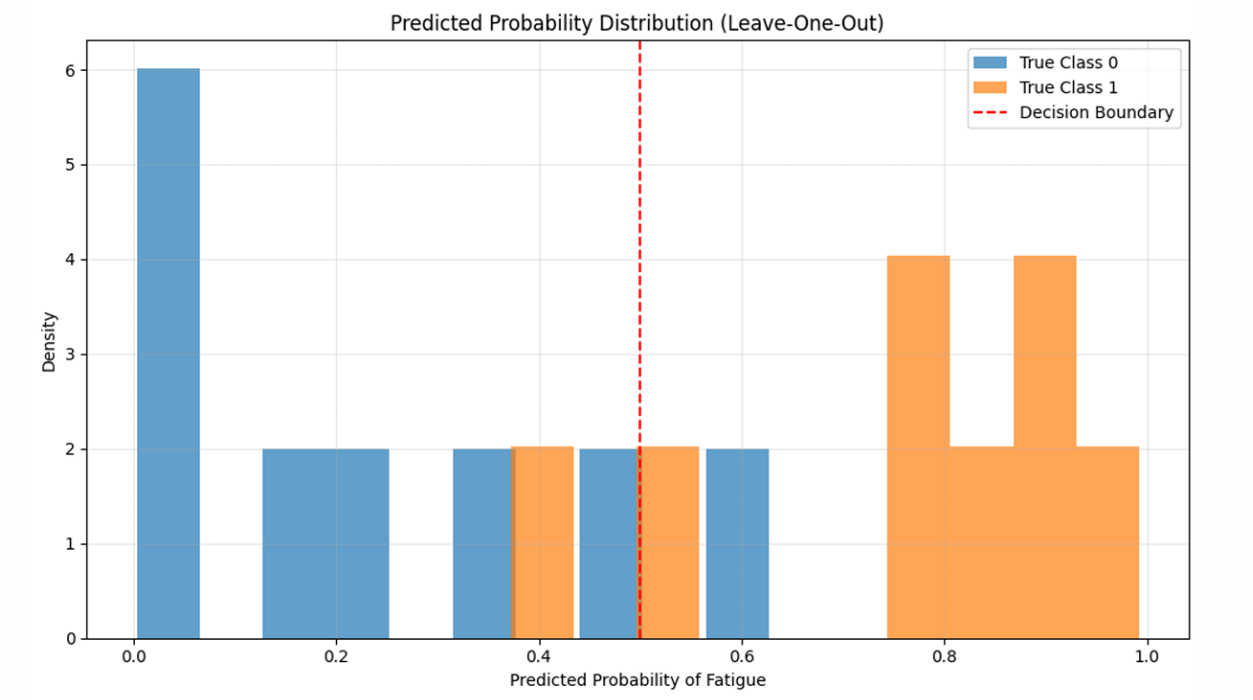


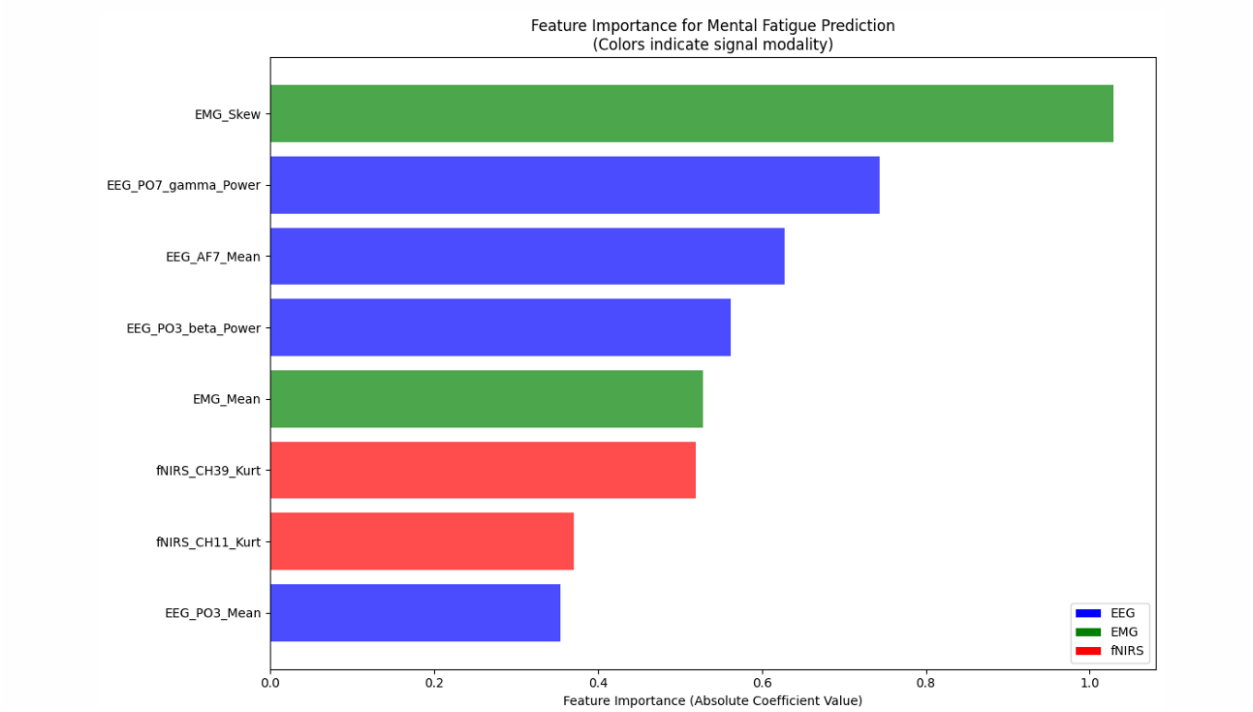

```
📈 COMPREHENSIVE SIGNAL SUMMARY - Participant 000
==========================================================================================
Modality   Duration (s)    Samples        Sampling Rate   Variance
------------------------------------------------------------------------------------------
EEG        45.0            23040          512             1.79e+02
ECG        41.0            10259          250             6.58e+04
EMG        41.0            10259          250             2.49e+03
fNIRS      45.0            720            16              1.28e-04
✅ BLOCK 4B: Enhanced signal visualization functions defined
```

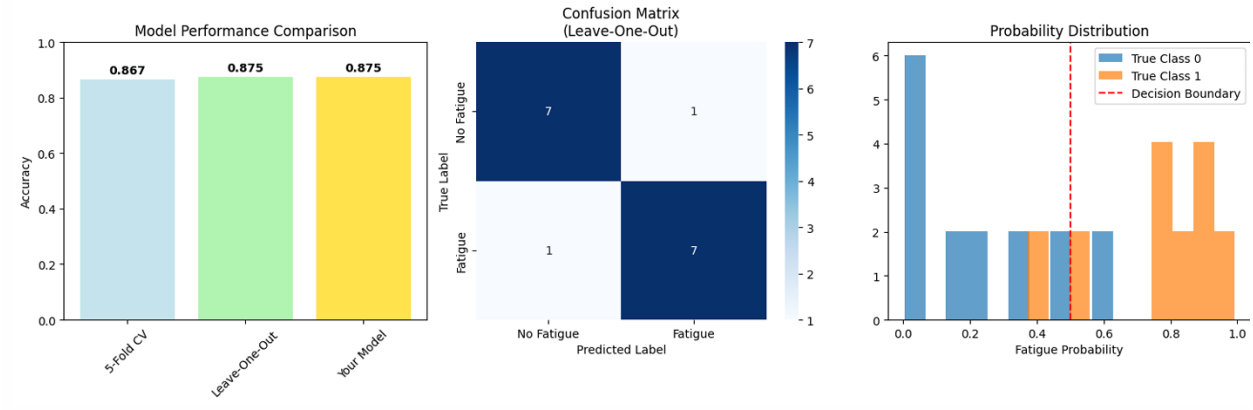

Feature Distributions by Class (First 12 Features)
EEG_AF7_Mean
EEG_AF7_STD
EEG_AF7_Var
EEG_AF7_Skew
EEG_AF7_Kurt
EEG_AF7_Med
EEG_AF7_Range
EEG_AF7_delta_Power
EEG_AF7_delta_RelPower
EEG_AF7_theta_Power
EEG_AF7_theta_RelPower
EEG_AF7_alpha_Power
Class 0
Class 1


Top 20 Most Important Features
0.030
0.025
0.020
0.015
0.010
0.005
0.000
EMG_Skew
EEG_F3_beta_Power
EEG_F3_theta_Power
EEG_AF8_Var
EEG_PO7_Range
ECG_HF
EEG_AF8_Mean
EEG_PO3_Kurt
EEG_PO8_gamma_RelPower
ECG_Var
EEG_F3_Kurt
EEG_F3_alpha_Power
EEG_PO3_beta_Power
fNIRS_CH31_Kurt
EEG_PO3_Range
EEG_AF7_gamma_Power
EMG_Mean
EEG_F3_Mean
EEG_F3_STD
ECG_STD


Confusion Matrix - Leave-One-Out Validation
No Fatigue
Fatigue
True Label
7
1
1
7
No Fatigue
Fatigue
Predicted Label
7
6
5
4
3
2
1


Predicted Probability Distribution (Leave-One-Out)
True Class 0
True Class 1
Decision Boundary
Density
Predicted Probability of Fatigue
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0


Feature Importance for Mental Fatigue Prediction
(Colors indicate signal modality)
EMG_Skew
EEG_PO7_gamma_Power
EEG_AF7_Mean
EEG_PO3_beta_Power
EMG_Mean
fNIRS_CH39_Kurt
fNIRS_CH11_Kurt
EEG_PO3_Mean
EEG
EMG
fNIRS
0.0
0.2
0.4
0.6
0.8
1.0
Feature Importance (Absolute Coefficient Value)


Model Performance Comparison
0.867
0.875
0.875
Accuracy
5-Fold CV
Leave-One-Out
Your Model
Confusion Matrix
(Leave-One-Out)
No Fatigue
Fatigue
True Label
Predicted Label
7
1
1
7
Probability Distribution
True Class 0
True Class 1
Decision Boundary
Fatigue Probability

📋 QUICK SUMMARY TABLE:

Participant	True	Predicted	Probability	Status
000	1	1	0.992	✅ CORRECT
001	0	0	0.246	✅ CORRECT
002	1	1	0.541	✅ CORRECT
003	0	0	0.178	✅ CORRECT
004	1	1	0.789	✅ CORRECT
005	0	0	0.324	✅ CORRECT
006	1	1	0.907	✅ CORRECT
007	0	0	0.457	✅ CORRECT
008	1	1	0.829	✅ CORRECT
009	0	0	0.013	✅ CORRECT
010	1	0	0.373	❌ WRONG
011	0	0	0.003	✅ CORRECT
012	1	1	0.881	✅ CORRECT
013	0	0	0.038	✅ CORRECT
014	1	1	0.767	✅ CORRECT
015	0	1	0.627	❌ WRONG

==========

DETAILED INDIVIDUAL ANALYSIS REPORTS

==========

```
Initializing Offline Fatigue Analyzer with 87.5% accurate model...

============================================================
COMPREHENSIVE MENTAL FATIGUE ANALYSIS SYSTEM
============================================================
📊 COMPLETE GROUP FATIGUE ANALYSIS
==================================================
Total Participants: 16
Correct Predictions: 14/16
Overall Accuracy: 0.8750 (87.5%)
Leave-One-Out Validation: 0.8750 (87.5%)

🔍 MISCLASSIFIED PARTICIPANTS:
  Participant 010:
    True: Fatigue
    Predicted: No Fatigue
    Probability: 0.373
    Confidence: 0.627
  Participant 015:
    True: No Fatigue
    Predicted: Fatigue
    Probability: 0.627
    Confidence: 0.627

============================================================
DETAILED INDIVIDUAL REPORTS - ALL 16 PARTICIPANTS
============================================================
```

```
[1/16]
🧠 FATIGUE ANALYSIS REPORT - Participant 000
============================================================
True State: Fatigue
Predicted State: Fatigue
Fatigue Probability: 0.992
Prediction Confidence: 0.992
Emotion Quadrant: 2
Correct Prediction: ✅ YES

🚨 FATIGUE DETECTED - Recommendations:
  1. Take a 15-minute break from current tasks
  2. Practice deep breathing exercises (4-7-8 technique)
  3. Hydrate with water - dehydration affects cognitive performance
  4. Consider a short walk to improve circulation
  5. Monitor workload and prioritize tasks
  6. Ensure adequate sleep tonight (7-9 hours)
```

```
[2/16]
🧠 FATIGUE ANALYSIS REPORT - Participant 001
================================================
True State: No Fatigue
Predicted State: No Fatigue
Fatigue Probability: 0.246
Prediction Confidence: 0.754
Emotion Quadrant: 1
Correct Prediction: ✅ YES

✅ NO FATIGUE DETECTED - Maintenance Recommendat
  1. Continue current work-rest balance
  2. Maintain good posture and take micro-breaks
  3. Stay hydrated and have balanced nutrition
  4. Practice 5-minute mindfulness meditation
  5. Monitor energy levels proactively
  6. Maintain consistent sleep schedule
```

Reference :

1. Multimodal Fusion for Emotion Recognition

Title: A Comprehensive Review on Multimodal Emotion Recognition Using Deep Learning
Authors: K. R. A. et al.
Relevance to Your Work: This paper provides a comprehensive overview of why and how to combine multiple signals (like EEG, ECG, etc.) for emotion recognition. It justifies your core approach of moving beyond single-modality systems, discussing the strengths of different biosignals and fusion techniques, which directly supports your introduction and problem statement.
Link: **https://link.springer.com/article/10.1007/s42979-021-00675-9**

2. CNN + LSTM for Biosignal Analysis

Title: EEG-Based Emotion Recognition Using Convolutional Neural Network and Long Short-Term Memory
Authors: Y. Li, J. Huang, H. Zhou, N. Zhong
Relevance to Your Work: This is a foundational paper that demonstrates the effectiveness of the hybrid CNN-LSTM model for processing biosignal data. The CNN extracts spatial/spectral features, and the LSTM captures temporal dependencies, which is precisely the methodology you employed. It validates your model choice.
Link: **https://ieeexplore.ieee.org/document/9229567**

3. fNIRS for Cognitive State and Fatigue Detection

Title: A systematic review on the use of fNIRS in mental fatigue detection
Authors: M. A. H. R. et al.
Relevance to Your Work: Your project uses fNIRS to measure brain oxygenation related to fatigue. This systematic review consolidates evidence that fNIRS is a reliable tool for detecting mental fatigue, providing a strong reference for including this modality in your multimodal framework.
Link: **https://www.sciencedirect.com/science/article/pii/S1746809423001731**

4. Fatigue Detection from EEG and ECG

Title: Mental Fatigue Detection Using a Wearable, Consumer-Grade EEG Headset
Authors: J. E. K. et al.
Relevance to Your Work: This paper is highly relevant as it tackles fatigue detection using EEG, similar to your work. It discusses feature extraction (like band powers) and machine learning for classification, reinforcing the methods you used for the EEG and ECG components of your project.
Link: **https://ieeexplore.ieee.org/abstract/document/8954222**

5. A Foundational Dataset for Multimodal Emotion Research

Title: DEAP: A Dataset for Emotion Analysis using EEG, Physiological and Video Signals
Authors: S. Koelstra et al.
Relevance to Your Work: While you used a dataset from IEEE Dataport, the DEAP dataset is one of the most famous and widely used public datasets for multimodal emotion recognition. Citing it places your work within the established field of research and provides a benchmark for the type of data and labels (like valence-arousal) you are working with.
Link: **https://ieeexplore.ieee.org/document/5871728**